CPE Studio AI

Prototype 1 — Recording Tool · Stevens CPE

1. Plan

We discussed two ideas for replacing Riverside:

Idea 1 — Complete tool

Professors record their camera and screen, and the team edits and exports everything in the same place.

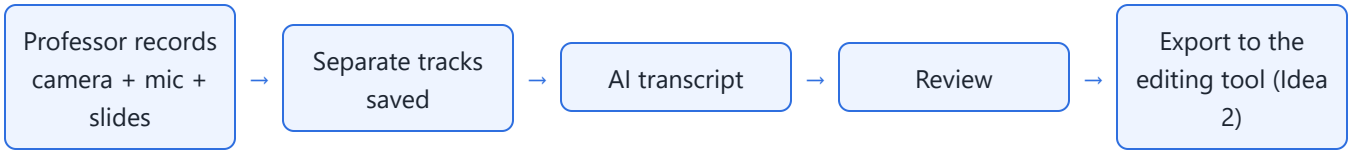
Prototype 1 — recording part

Idea 2 — Editing-only tool

Upload recordings, edit through the transcript, remove filler words and pauses, and export the videos.

Already ready

Decision: work in detail on Idea 1, building only the **recording part** together with the **transcript and the AI features Riverside has**. Editing is not part of this prototype; it is handled by the Idea 2 editing tool, which receives the recorded files and transcript.



2. Prototype 1 — Implemented features

Recording

Feature	Description
Camera + microphone	Up to 1080p / 30 fps, with noise suppression and echo cancellation. The camera and microphone permission prompt appears as soon as the page opens.
Screen / slide sharing	Shares the entire screen, a window or a browser tab.
Screen audio	Records sound from videos played during the lecture, with a live indicator showing whether screen audio is connected.
Separate synchronized tracks	Records three files (camera + mic, screen, audio only) with their start offsets saved, so they stay in sync.
Camera / microphone on-off	The professor can switch the camera or microphone off and on at any time. The recording continues without breaking, and each toggle is logged.
Recording controls	Start, pause / resume and stop, with a timer and level meters for the microphone and screen audio.

Device settings	Choose the camera and microphone. Clear messages appear when a device is busy or blocked, with a Retry button.
Auto-save & recovery	Saves every 4 seconds while recording. An interrupted session can still be recovered from Recordings.

AI features

Feature	Description
Live captions	Speech is shown as text while the professor is recording.
AI transcript	Whisper speech-to-text with a timestamp for every word.
Interactive transcript	Click a word to jump to that moment in the video. The current word is highlighted during playback. Double-click a word to correct its spelling.

Review & export

Feature	Description
Playback	Camera and slides play in sync, with a switch between the Camera view and the Slides view.
Recordings list	Open or delete saved recordings.
Export	Camera, screen and audio files, the transcript (.srt, .txt, .json) and a project manifest for the editing tool. Includes a single Download everything button.

3. Tech stack

Layer	Technology
Interface	HTML, CSS, JavaScript (runs in Chrome / Edge)
Camera, microphone & screen capture	Browser Media APIs: getUserMedia, getDisplayMedia
Recording	MediaRecorder (WebM video and audio)
Camera / mic on-off processing	Insertable Streams (black frames) and Web Audio API (silence)
Local storage & recovery	IndexedDB in the browser
Live captions	Web Speech API (built into Chrome / Edge)
AI transcript	Whisper API: Groq *whisper-large-v3-turbo* (free tier) or OpenAI *whisper-1*
Local server	PowerShell script started with *start.bat*; no installation required

4. Future features (local machine)

Feature	Benefit
Offline AI transcript	Run Whisper on the local machine (GPU) so lecture audio never leaves the computer and no API key is needed.
Save to a local folder	Write recordings directly into a chosen folder on the computer instead of browser storage.
MP4 export	Convert recordings to MP4 on the machine (FFmpeg) for wider compatibility.
Long-lecture transcription	Split long audio automatically so lectures of any length can be transcribed.
Audio enhancement	Clean up noise and echo after recording, similar to Riverside's audio enhancement.
AI summary & chapters	Generate a lecture summary, chapter markers and key terms from the transcript.
Translated captions	Create captions in other languages from the transcript.
Slide-change markers	Detect when the slide changes and mark it on the timeline to make editing easier.
Pre-recording check	Test the camera, microphone, screen audio and free disk space before starting.
Direct handoff to the editing tool	Open a finished recording in the Idea 2 editing tool with one click.